Trajectory Estimation for a Venus Aerobot Balloon during Deep Space Network Occlusion

Giorgio Morra[a,b,*], Francesco Braghin[a], Marco B. Quadrelli[b], Federico Rossi[b]

[a] *Department of Mechanical Engineering, Politecnico di Milano, Milano, Italy*
[b] *Jet Propulsion Laboratory, California Institute of Technology, Pasadena, CA, USA*

Abstract

At Venus, absolute position is normally recovered through radiometric tracking by the Deep Space Network. When a balloon drifts onto the far side of the planet, Venus occults the line of sight: a blackout of up to 65 h, during which about 14 508 km are flown with no contact. This work estimates the trajectory of a Venus aerobot over the sunlit part of the occlusion, about 46 h before the Sun sets, by fusing an IMU, a Sun sensor, a radar altimeter and a barometer. A Multiplicative Extended Kalman Filter tracks the attitude and a particle filter tracks the position, coupled in a cascade that re-injects every transferred uncertainty as an inflated measurement covariance. Over the sunlit arc, Monte Carlo campaigns of 100 runs give a position RMSE of 26.7 km, never worse than 34.3 km, within the 50 km requirement. A single Sun channel on the position estimation raises convergence from about 60% to 98–99%, while the accuracy is set by the onboard atmospheric model rather than by the quality of the Sun sensor.

Keywords: Venus aerobot, Autonomous navigation, Multiplicative extended Kalman filter, Particle filter, DSN blackout, Sensor fusion

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*Corresponding author
Email address: giorgiomorra1@outlook.com (Giorgio Morra)

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1. Introduction

1.1. Venus and the Balloon Mission

Venus is often called Earth's sister planet: it formed from similar building blocks, at a similar distance from the Sun, and it is almost the same size and mass. Its evolution, however, diverged completely. A slowly rotating early Venus may have sustained temperate surface conditions, and possibly a surface ocean [1]; at some point it lost its water and entered a runaway greenhouse state, and today its surface sits at roughly 740 K under a pressure near 92 bar [2]. How two such similar planets ended up so different is an open question of comparative planetology, all the more relevant since Venus-sized worlds are among the most common found by Kepler [3].

Volcanism is tied to the same questions. The surface is almost entirely volcanic plain, with remarkably few impact craters for a planet of its size

and age [4], and whether its resurfacing was a single global pulse or is still ongoing today remains unresolved, although changes observed in repeated Magellan radar images point towards present-day activity [5]. A concept developed at JPL proposes to study eruptions as they happen with altitude-controlled balloons carrying infrasound microbarometers that manoeuvre to within 50 km of a detected event [6]. Such observations cannot be made from orbit, because a deck of sulfuric-acid clouds hides the surface from any instrument working at visible wavelengths. Inside the atmosphere, however, between 50 km and 60 km of altitude, temperature and pressure approach those at the Earth's surface, about 303 K and 0.52 bar near 55 km [7], so that an aerial platform can operate there.

In June 1985 the Soviet VEGA 1 and VEGA 2 missions released two helium balloons at about 54 km, the first and to date only aerial vehicles to have operated at Venus. Tracked for about 46 h by radio telescopes, they returned the first in-situ measurements of the cloud-layer winds [8], but with no altitude control they were pure passive tracers of the flow. The variable-altitude aerobot developed at JPL is a direct answer to that limitation [9]: helium is pumped between an outer zero-pressure and an inner super-pressure envelope (Fig. 1) to control altitude between 52 km and 62 km, and prototypes have been flown on Earth [10]. The vehicle has no propulsion: it is carried around the planet by the super-rotating atmosphere, which at cloud level circulates in a few days against the 243-day rotation of the solid body [11, 2], and altitude is the only degree of freedom it actively controls.

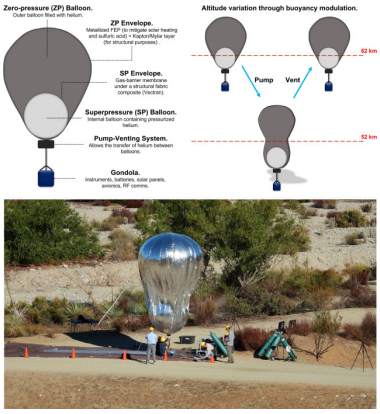


Figure 1: The two-part variable-altitude aerobot: helium pumped into the inner envelope makes the vehicle descend, venting it makes it rise (52–62 km). Adapted from Hall et al. [9].

1.2. The Localization Problem

Every measurement acquired by the balloon is scientifically useful only if it can be attributed to a known position. Since the balloon has no propulsion, its trajectory is close to the time integral of the local wind, and if that wind were known there would be no estimation problem. The wind, however, is not deterministic at the scales and horizons that matter: superimposed on the mean circulation is a hierarchy of disturbances, from planetary-scale waves [12, 13, 14] and low-frequency variability of the jet [15] to meso-scale gravity waves and convection [16] and orographic waves over the highlands [17, 18]. In Fig. 2, ten trajectories are propagated from the same initial state on the perturbed version of the Venus Climate Database [7] and compared with the trajectory propagated on the mean fields of the same database: their distance from this nominal prediction grows almost linearly, at about $14\,\mathrm{km\,h^{-1}}$.

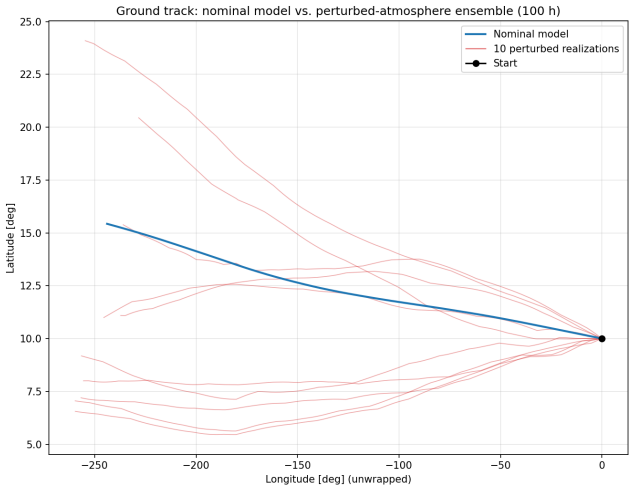


Figure 2: Ground track on the mean-field atmosphere vs. ten realisations on the perturbed atmosphere with additional disturbances, from the same initial state: the error grows at about 14 km/h, roughly 900 km at the end of the 65 h blackout.

On the Earth-facing side of the planet the problem is solved by radiometric tracking from NASA's Deep Space Network (DSN), and this phase is not addressed here. On the far side, however, Venus occults the line of sight: the blackout lasts almost 65 h, about 14 508 km of ground track with no contact and no satellite navigation to fall back on. An open-loop extrapolation would end it with an error of order 900 km, eighteen times the 50 km required to close on a volcanic event [6], adopted here as the minimum requirement on the reconstructed position.

The far-side window is not homogeneous: the Sun sets once, about 46 h after the link is lost, and does not rise again before the DSN reappears. The window thus splits into a sunlit arc of about 46 h, in which all four onboard sensors (inertial unit, Sun sensor, radar altimeter, barometer) are available, and a terminal eclipse of about 19 h, in which the Sun sensor cannot be used. This paper addresses the sunlit arc, about 46 h and 10 000 km without any absolute fix; all the accuracy figures of Section 5 refer to it, and the eclipse is discussed in Section 6 as the open problem.

2. State of the Art

During the blackout, position information comes from terrain- and map-referenced navigation: the altimeter and the barometer are matched against the known topography [19] and pressure field [7]. Because distinct locations produce identical readings, the resulting likelihood is multi-modal, which a Kalman-type filter [20, 21] linearised about a single mean cannot represent. Particle filters [22, 23] are the natural remedy, and their use for terrain navigation and map matching is well established [24].

Two published architectures address the Venus balloon specifically (Table 1). Ploen et al. [25] analyse the covariance growth of a balloon navigating on an inertial unit alone, with periodic orbiter updates, and propose bias-compensation strategies including a mechanically rotated IMU, over a ten-minute horizon. Cottrill and Gu [26] correct the drift of an inertial navigation system by map matching altitude and pressure in a particle filter with fifteen states and 5000 particles, and report a position RMSE of 47.95 km over 120 Monte Carlo trials of one hour each.

The main limitation of both lies in the duration of the simulations. In one hour the balloon drifts by little more than 200 km, whereas the sunlit arc considered here lasts about 46 h, one to two orders of magnitude longer. Over such short horizons the errors that dominate a long autonomous flight (bias drift, the lock onto a false terrain match) have no time to develop, so neither study shows that its architecture can bridge the far side. This work further adds a Sun sensor, the only measurement of absolute longitude, without which four runs in ten diverge (Section 4), and estimates the attitude in a separate Multiplicative Extended Kalman Filter (MEKF) [27, 28] coupled in cascade with the particle filter.

Table 1: Navigation architectures for a Venus balloon, compared with this work.

	Ploen et al. [25]	Cottrill and Gu [26]	This work
Approach	Bias-compensated inertial navigation, rotated IMU	Inertial navigation, map matching and particle filter, 15 states, 5000 particles	Coupled MEKF and particle filter, 9-state attitude error and 6-state position, 1000 particles
Measurements	IMU only, with orbiter updates	IMU, altimeter, barometer	IMU, altimeter, barometer, Sun sensor
Maps	None	VCD, Magellan topography	VCD, Magellan topography
Horizon	10 min (covariance study)	1 h, 120 Monte Carlo trials	46 h sunlit arc of the 65 h far-side window, 100 Monte Carlo trials
Reported error	Not reported	47.95 km position RMSE	26.7 km position RMSE (sunlit arc)

3. Dynamic Model and Sensor Simulation

3.1. Reference Frames, Time Systems and Simulation Environment

All planetary geometry and ephemerides are obtained from NASA's NAIF SPICE toolkit [29]. Four reference frames are used (Fig. 3):

- **Venus-centred inertial (VCI, J2000)**, non-rotating with respect to the stars, in which the positions of the Sun and of the Earth are computed.
- **Venus-centred, Venus-fixed (VCVF, IAU_VENUS)**, corotating with the planet at $\boldsymbol{\omega}_p = [0, 0, \Omega]^\top$, $\Omega = -2.99 \times 10^{-7}\,\mathrm{rad\,s^{-1}}$ (retrograde); the atmospheric database is read and the translational dynamics are written in this frame, and its rotation from J2000 is retrieved from SPICE.
- **East–North–Up (ENU)**, the local horizontal frame at the balloon, in which the winds are expressed.
- **Body frame (B)**, attached to the vehicle at its centre of mass, in which the inertial unit, the Sun sensor and the altimeter measure; its

orientation with respect to VCVF is the unit quaternion $\mathbf{q}$, in scalar-last convention [28], with direction-cosine matrix $\mathbf{R} = \mathbf{R}_{B\to VCVF}(\mathbf{q})$.

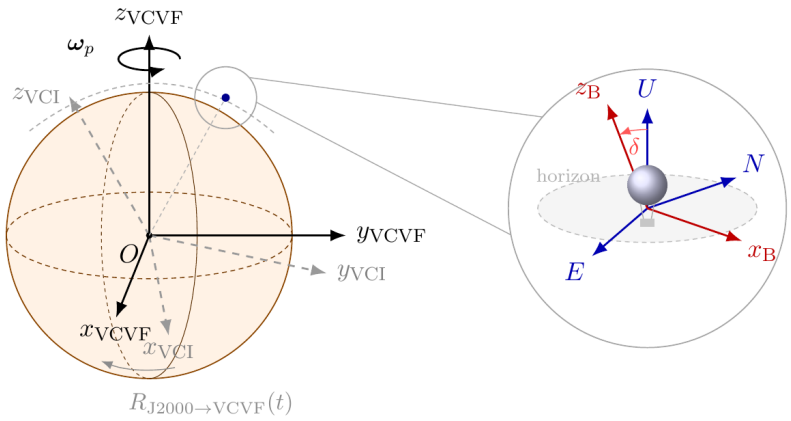


Figure 3: Reference frames. Left: inertial VCI/J2000 and planet-fixed VCVF. Right: local ENU frame and body frame at the vehicle, tilted by δ from the local vertical.

Three time systems are used:

- **Coordinated Universal Time (UTC)**, in which the start epoch is defined;
- **Ephemeris Time (ET)**, the seconds passed since the J2000 epoch, used by SPICE;
- **Venus local solar time (LT)**, which indexes the Venus Climate Database [7]; it is measured in Venus hours, each one twenty-fourth of the 116.75 h solar day, and is shifted with the longitude of the vehicle.

Because the problem is defined by the loss of the DSN link, the scenario is placed in a configuration that a real mission would actually fly, close to an inferior conjunction, when Venus, the Sun and the Earth are nearly aligned and the Earth–Venus distance is at its minimum: the start epoch of Table 2 falls about five weeks after the inferior conjunction of 10 August 2031. The balloon is released on the Earth-facing hemisphere, where the DSN is available, and the zonal wind carries it onto the far side, where the 65 h autonomous window begins: the Sun stays visible for its first 46 h, and neither the Earth nor the Sun is in view for the last 19 h. The ground truth is propagated on the perturbed version of the Venus Climate Database, whereas the estimator only knows its mean fields, so that a persistent modelling error is present in every result.

Table 2: Scenario and initial conditions, common to all simulations.

Quantity	Value
Start epoch	2031-09-13 10:00:00 UTC
Duration / integration step	100 h / 0.1 s
Float altitude	≈58 km
Initial longitude	0°
Latitude band	−20° to 20°

3.2. Balloon 6-DoF Dynamics

The same model generates the ground truth and the measurements, with the parameters of the B2 aerobot design [9]. For the translational motion the balloon is a point mass whose gas is in thermal and pressure equilibrium with the atmosphere, the envelope is a sphere, and the helium ratio between the two envelopes is held constant, so that altitude control is not commanded. The volume is $V = n_{ZP}RT/(PM_{He}) + V_{SP}$, with n_{ZP} the helium in the zero-pressure envelope and V_{SP} the constant super-pressure volume, the reference area is $A_{ref} = \pi(3V/4\pi)^{2/3}$, and the total mass $m_{tot} = m_{dry} + \frac{1}{2}\rho V$ includes the added mass of the displaced atmosphere. Gravity $\mathbf{F}_{grav} = -m_{dry}g_0\hat{\mathbf{r}}$, buoyancy $\mathbf{F}_{buoy} = \rho V g_0 \hat{\mathbf{r}}$ and drag $\mathbf{F}_{drag} = -\frac{1}{2}\rho C_d A_{ref}\|\mathbf{v}_{rel}\|\mathbf{v}_{rel}$, with $\mathbf{v}_{rel} = \mathbf{v} - \mathbf{W}$ the velocity relative to the database wind, act on the vehicle, and since VCVF rotates the acceleration includes the centrifugal and Coriolis terms.

For the rotational motion, envelope, tether and gondola form a single rigid body. The centres of buoyancy and pressure lie 8 m above the centre of mass, so the external torque $\mathbf{T}_{ext} = \mathbf{r}_{CB} \times \mathbf{R}^\top \mathbf{F}_{buoy} + \mathbf{r}_{CP} \times \mathbf{R}^\top \mathbf{F}_{drag} - \mathbf{C}_{damp}\boldsymbol{\omega}_B + \mathbf{T}_{dist}$ contains a pendulum-like restoring moment about the local vertical, a linear damping, and a first-order Gauss–Markov disturbance $\dot{\mathbf{T}}_{dist} = \boldsymbol{\Pi}\mathbf{T}_{dist} + \boldsymbol{\eta}_T/\Delta t$ that represents the aerodynamic disturbances on the envelope [25]. Euler's equation uses the inertial angular velocity $\boldsymbol{\omega}_B^{VCI} = \boldsymbol{\omega}_B + \mathbf{R}^\top\boldsymbol{\omega}_p$. The state $\mathbf{X} = [\mathbf{r}^\top, \mathbf{v}^\top, \mathbf{q}^\top, \boldsymbol{\omega}_B^\top]^\top \in \mathbb{R}^{13}$ then evolves as

$$
\begin{cases}
\dot{\mathbf{r}} = \mathbf{v}, \\
\dot{\mathbf{v}} = \dfrac{\mathbf{F}_{grav} + \mathbf{F}_{buoy} + \mathbf{F}_{drag}}{m_{tot}} - \boldsymbol{\omega}_p \times (\boldsymbol{\omega}_p \times \mathbf{r}) - 2\,\boldsymbol{\omega}_p \times \mathbf{v}, \\
\dot{\mathbf{q}} = \frac{1}{2}\,\boldsymbol{\Omega}(\boldsymbol{\omega}_B)\,\mathbf{q}, \qquad \boldsymbol{\Omega}(\boldsymbol{\omega}) = \begin{bmatrix} -[\boldsymbol{\omega}\times] & \boldsymbol{\omega} \\ -\boldsymbol{\omega}^\top & 0 \end{bmatrix}, \\
\dot{\boldsymbol{\omega}}_B = \mathbf{J}_B^{-1}\left(\mathbf{T}_{ext} - \boldsymbol{\omega}_B^{VCI} \times \mathbf{J}_B\,\boldsymbol{\omega}_B^{VCI}\right).
\end{cases}
\tag{1}
$$

The result is a damped physical pendulum: roll and pitch stay within about 1° of the local vertical, while yaw, with no restoring moment, evolves as a random walk.

3.3. Sun Sensor

The Sun sensor is the only absolute, sky-referenced measurement in the suite, and the only instrument whose realisability at Venus is genuinely open: no previous study has addressed the use of a Sun sensor on a Venus aerobot, so its feasibility was assessed first. An optical or near-infrared sensor was considered first and turned out not to be feasible. The sulfuric-acid cloud deck between 50 km and 70 km scatters in the Mie regime with an optical depth of order 25 [30]; by the Beer–Lambert law, $I(h)/I_0 = \exp(-\tau(h)/\cos\theta_z)$, and assuming τ grows linearly from 0 at 70 km to 25 at 50 km, the direct beam is already negligible at 66 km, well above the float altitude. The measurement was therefore moved to the microwave band. From the atmospheric weighting functions of Akins et al. [31], the usable band is between roughly 1 GHz and 18 GHz, where the absorption below the balloon screens the thermal emission of the hot surface while the atmosphere above it is transparent to the solar signal. The direction of the Sun can be recovered by interferometry between small patch antennas: for a baseline b and wavelength λ_{RF}, a source at angle θ from the baseline normal produces a phase difference $\Delta\phi = 2\pi b \sin\theta/\lambda_{RF}$, and a second, orthogonal baseline resolves the full direction. This work does not go into the detail of the hardware or of the antenna configuration: it only establishes that a Sun measurement is physically available at float altitude, while the accuracy such an instrument might deliver is covered by the sensitivity analysis of Section 5.

In the simulator the Sun direction from SPICE is rotated into the body frame, $\mathbf{s}_B = \mathbf{R}^\top \mathbf{s}_{VCVF}$, and the sensor reports $az = \mathrm{atan2}(\hat{s}_y, \hat{s}_x)$ and $el = \arcsin(\hat{s}_z)$ of the unit vector $\hat{\mathbf{s}}$. The measurement makes the yaw observable and, being a function of absolute latitude and longitude, also feeds the position filter (Section 4).

3.4. Radar Altimeter

The altimeter ties the vehicle to the Magellan topography of Venus [19]: over highland massifs such as Aphrodite Terra the measurement is strongly informative on horizontal position, over the nearly flat plains only weakly. The instrument points along the body $-Z$ axis, and the measurement is simulated by ray casting (Fig. 4): the ray $\mathbf{n} = \mathbf{R}\,[0, 0, -1]^\top$ is intersected with the Venus reference ellipsoid through SPICE, the terrain elevation at the hit point is read from the topography map, and the slant range is the distance from the vehicle to that surface point. The measurement therefore accounts for the attitude of the balloon, since any tilt away from the local vertical lengthens the range and moves the footprint on the ground.

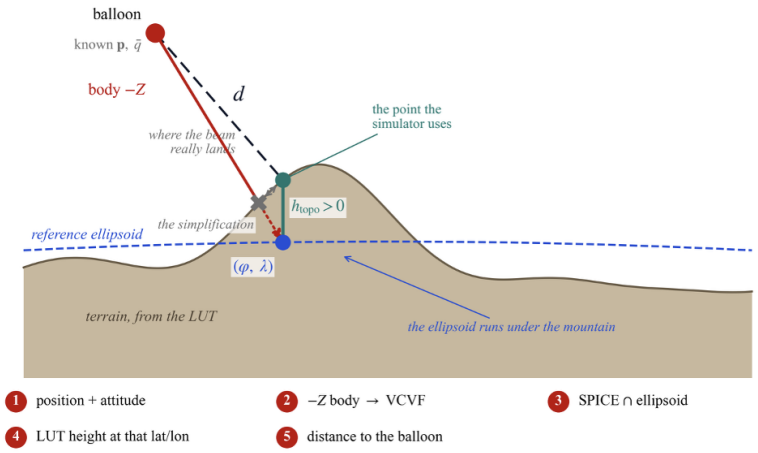


Figure 4: Altimeter ray casting along the body $-Z$ axis against the reference ellipsoid; the terrain elevation at the hit point sets the slant range.

Remark on the measurement set. Table 3 summarises the four onboard sensors and their error models. The inertial unit is a MEMS-grade unit of the class considered for Venus balloons [25], with the Case 3 specifications of Cottrill and Gu [26]: each axis of the accelerometer, which measures the specific force in body axes, and of the gyroscope carries a turn-on bias, a bias random walk and white noise. The other sensors are corrupted by additive zero-mean Gaussian noise; for the Sun sensor, whose hardware is not designed in detail, two values larger than the baseline are also tested, to verify that the approach does not depend on this optimistic figure.

Table 3: Onboard measurement set and error models.

Sensor	Output	Error model	Constrains
Gyroscope	body rate	turn-on bias $0.15\,^\circ\,\mathrm{h}^{-1}$, bias random walk $0.015\,^\circ\,\mathrm{h}^{-1}\,\mathrm{s}^{-0.5}$, angle random walk $0.01\,^\circ/\sqrt{\mathrm{h}}$	attitude
Accelerometer	specific force	turn-on bias $300\,\mu\mathrm{g}$, bias random walk $30\,\mu\mathrm{g}\,\mathrm{s}^{-0.5}$, white noise $0.02\,\mathrm{m}\,\mathrm{s}^{-1}/\sqrt{\mathrm{h}}$	pitch, roll
Sun sensor	azimuth, elevation	Gaussian, $\sigma_s = 1 \times 10^{-4\circ}$ (baseline), $1 \times 10^{-2\circ}$, $1 \times 10^{-1\circ}$	yaw, tilt, position
Altimeter	slant range	Gaussian, $\sigma_{alt} = 5\,\mathrm{m}$	horizontal position
Barometer	pressure	Gaussian, $\sigma_{bar} = 40\,\mathrm{Pa}$	altitude

4. Estimation Architecture

4.1. The Cascade Architecture

Position is estimated by a particle filter and attitude by a Multiplicative Extended Kalman Filter (MEKF), coupled in cascade (Fig. 5).

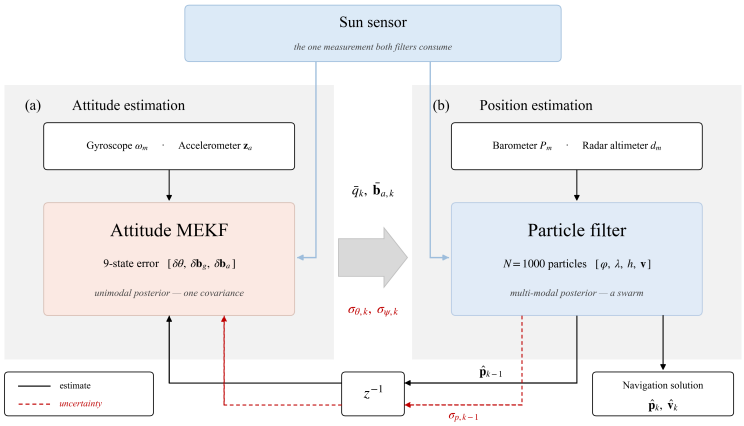


Figure 5: Estimation architecture: the attitude MEKF (a), carrying a nine-state error and a single covariance, and the position particle filter (b), carrying 1000 particles over position and velocity. Solid arrows are estimates, dashed red arrows are uncertainties.

Within each epoch k:

1. the local vertical and the Sun direction are evaluated at the position estimate produced by the particle filter at epoch $k-1$;
2. the MEKF runs its predict, correct and reset cycle on the gyroscope, the accelerometer and the Sun sensor;
3. the quaternion, the accelerometer bias and the attitude uncertainty are handed to the particle filter;
4. the particle filter propagates the swarm and re-weights it against the barometer, the altimeter and the Sun angles;
5. the resulting position estimate and its spread are stored as the references of step 1 at the next epoch.

4.2. Attitude Estimation: the Multiplicative Extended Kalman Filter

The MEKF [27] is the state-of-the-art variant of the Kalman filter for quaternion-based attitude estimation. It propagates with the exact nonlinear kinematics a nominal state $\bar{\mathbf{X}} = [\bar{\mathbf{q}}^\top, \bar{\mathbf{b}}_g^\top, \bar{\mathbf{b}}_a^\top]^\top$, the quaternion and the gyroscope and accelerometer biases, and estimates the Gaussian error state $\delta\mathbf{X} = [\delta\boldsymbol{\theta}^\top, \delta\mathbf{b}_g^\top, \delta\mathbf{b}_a^\top]^\top \in \mathbb{R}^9$, with covariance $\mathbf{P}$, where $\mathbf{q}^{true} = \bar{\mathbf{q}} \otimes \mathbf{q}\{\delta\boldsymbol{\theta}\}$ and $\mathbf{q}\{\cdot\}$ maps a rotation vector to a unit quaternion [28].

Prediction. With the bias-corrected gyroscope rate $\boldsymbol{\omega}_k = \boldsymbol{\omega}_{m,k} - \bar{\mathbf{b}}_g$, the nominal quaternion is advanced as $\bar{\mathbf{q}}_k = \bar{\mathbf{q}}_{k-1} \otimes \mathbf{q}\{\boldsymbol{\omega}_k \Delta t\}$ and the nominal biases are held constant. The error state evolves as $\dot{\delta\boldsymbol{\theta}} = -[\boldsymbol{\omega}\times]\delta\boldsymbol{\theta} - \delta\mathbf{b}_g - \mathbf{n}_g$, with the biases as random walks, so that

$$\mathbf{P}_k^- = \mathbf{F}\mathbf{P}_{k-1}\mathbf{F}^\top + \mathbf{Q}, \qquad \mathbf{F} = \begin{bmatrix} \mathbf{I}_3 - \Delta t[\boldsymbol{\omega}\times] & -\Delta t\,\mathbf{I}_3 & \mathbf{0}_3 \\ \mathbf{0}_3 & \mathbf{I}_3 & \mathbf{0}_3 \\ \mathbf{0}_3 & \mathbf{0}_3 & \mathbf{I}_3 \end{bmatrix}. \tag{2}$$

The process-noise covariance is obtained with the Van Loan construction [32], integrating the noise together with the dynamics over the step. Since the dynamics matrix of the attitude and gyro-bias pair is nilpotent, its transition matrix is exactly $\boldsymbol{\Phi}(\tau) = \mathbf{I}_6 + \mathbf{A}\tau$, and integrating $\boldsymbol{\Phi}\,\mathrm{blkdiag}(\sigma_g^2\mathbf{I}_3, \sigma_{bg}^2\mathbf{I}_3)\,\boldsymbol{\Phi}^\top$ over $[0, \Delta t]$ gives

$$\mathbf{Q} = \begin{bmatrix} (\sigma_g^2\Delta t + \frac{1}{3}\sigma_{bg}^2\Delta t^3)\,\mathbf{I}_3 & -\frac{1}{2}\sigma_{bg}^2\Delta t^2\,\mathbf{I}_3 & \mathbf{0}_3 \\ -\frac{1}{2}\sigma_{bg}^2\Delta t^2\,\mathbf{I}_3 & \sigma_{bg}^2\Delta t\,\mathbf{I}_3 & \mathbf{0}_3 \\ \mathbf{0}_3 & \mathbf{0}_3 & \sigma_{ba}^2\Delta t\,\mathbf{I}_3 \end{bmatrix}, \tag{3}$$

with σ_g the gyroscope white noise and σ_{bg}, σ_{ba} the bias random-walk intensities of Table 3.

Measurement models. The accelerometer measures gravity along the local upversor $\hat{\mathbf{u}}(\mathbf{p})$, and the Sun sensor the Sun direction $\hat{\mathbf{s}}_{VCVF}(\mathbf{p})$, both resolved in body axes, $\hat{\mathbf{s}}_B = \mathbf{R}(\bar{\mathbf{q}})^\top \hat{\mathbf{s}}_{VCVF}(\mathbf{p})$:

$$\hat{\mathbf{z}}_a = \hat{\mathbf{g}}_B + \bar{\mathbf{b}}_a, \quad \hat{\mathbf{g}}_B = g_0\mathbf{R}(\bar{\mathbf{q}})^\top\hat{\mathbf{u}}(\mathbf{p}), \qquad \mathbf{H}_a = \begin{bmatrix}[\hat{\mathbf{g}}_B\times] & \mathbf{0}_3 & \mathbf{I}_3\end{bmatrix}, \tag{4}$$

$$\hat{\mathbf{z}}_s = \begin{bmatrix} \mathrm{atan2}(\hat{s}_{B,y}, \hat{s}_{B,x}) \\ \arcsin \hat{s}_{B,z} \end{bmatrix}, \qquad \mathbf{H}_s = \begin{bmatrix}\mathbf{J}_{ae}[\hat{\mathbf{s}}_B\times] & \mathbf{0}_{2\times3} & \mathbf{0}_{2\times3}\end{bmatrix}, \tag{5}$$

with $\mathbf{J}_{ae} = \partial(az, el)/\partial\hat{\mathbf{s}}_B$, whose azimuth row scales as $1/\cos(el)$, and noise covariances $\mathbf{R}_a = \mathrm{diag}(\sigma_w^2 + \sigma_{k,i}^2)$, where $\sigma_{k,i}$ accounts for the kinematic acceleration of the swinging vehicle, and $\mathbf{R}_s = \sigma_s^2\mathbf{I}_2$. Each correction computes $\mathbf{K} = \mathbf{P}\mathbf{H}^\top(\mathbf{H}\mathbf{P}\mathbf{H}^\top + \mathbf{R})^{-1}$ and $\delta\hat{\mathbf{X}} = \mathbf{K}(\mathbf{z} - \hat{\mathbf{z}})$, folds the error into the nominal state, $\bar{\mathbf{q}} \leftarrow \bar{\mathbf{q}} \otimes \mathbf{q}\{\delta\hat{\boldsymbol{\theta}}\}$ and $\bar{\mathbf{b}}_\bullet \leftarrow \bar{\mathbf{b}}_\bullet + \delta\hat{\mathbf{b}}_\bullet$, updates $\mathbf{P}$ in Joseph form, and resets the error state to zero.

4.3. Position Estimation: the Particle Filter

The terrain-relative likelihood is multi-modal in horizontal position, which no single Gaussian can represent. The particle filter [22, 23] represents the posterior by $N = 1000$ weighted samples, each a hypothesis on the translational state $\mathbf{X}^i = \{\mathbf{p}^i, \mathbf{v}^i\}$, $\mathbf{p}^i = [\varphi^i, \lambda^i, h^i]$. In this way, several possible

positions of the balloon can be kept at the same time, until the measurements make one of them clearly more likely than the others (Fig. 6).

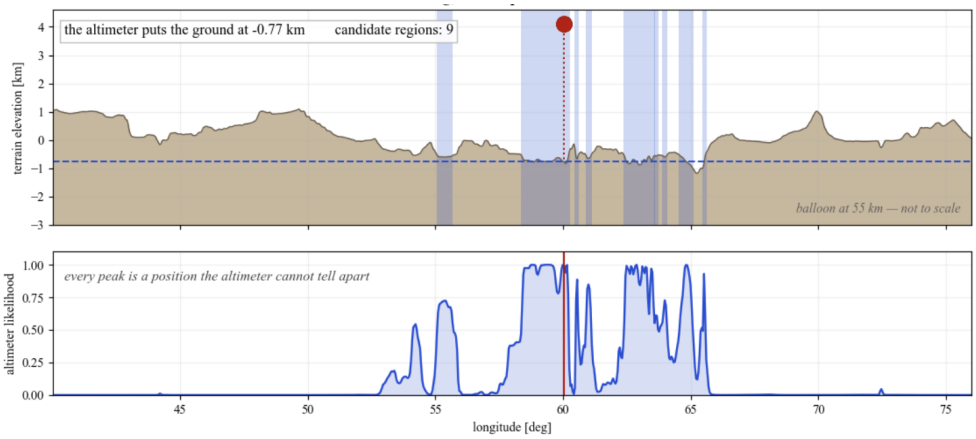


Figure 6: Multimodal posterior represented by the particle swarm: competing location hypotheses coexist as clusters until the measurements single one out.

Propagation. Each particle is propagated through the translational dynamics of Eq. (1), pushed by the wind, with the atmosphere $(P^i, \rho^i, T^i, \mathbf{W}^i)$ read at its own position from the nominal, mean fields of the Venus Climate Database [7]. Volume, reference area, mass and forces are evaluated per particle, and

$$\mathbf{a}^i = \frac{\mathbf{F}^i_{grav} + \mathbf{F}^i_{buoy} + \mathbf{F}^i_{drag}}{m^i_{tot}} - \boldsymbol{\omega}_p \times (\boldsymbol{\omega}_p \times \mathbf{r}^i) - 2\,\boldsymbol{\omega}_p \times \mathbf{v}^i + \boldsymbol{\eta}^i_a \qquad (6)$$

is integrated with an Euler step, the process noise $\boldsymbol{\eta}^i_a$ keeping the swarm from collapsing onto a single trajectory. Every particle thus flies the nominal atmosphere while the truth flies the perturbed one, and this discrepancy is the dominant model error of the estimator.

Measurement update. The barometer innovation is a direct map comparison, $\nu^i_{bar} = P_m - P_{LUT}(\mathbf{p}^i, t)$. The altimeter innovation stretches the vertical terrain clearance $\hat{h}^i_\perp$ of the map by the tilt of the body $-Z$ axis $\mathbf{n} = -\mathbf{R}(\bar{\mathbf{q}})\hat{\mathbf{z}}$,

$$\nu^i_{alt} = d_m - \frac{\hat{h}^i_\perp}{\cos\gamma^i}, \qquad \cos\gamma^i = -\left[\mathbf{R}_{VCVF\to ENU}(\mathbf{p}^i)\,\mathbf{n}\right]_U . \qquad (7)$$

The Sun sensor is informative for position only in the local horizon frame. The measured Sun direction and the specific force, de-biased with the MEKF

estimate, are rotated into VCVF with the shared attitude,

$$\hat{\mathbf{u}}_{meas} = \mathbf{R}(\bar{\mathbf{q}})\frac{\mathbf{z}_a - \bar{\mathbf{b}}_a}{\|\mathbf{z}_a - \bar{\mathbf{b}}_a\|}, \qquad \hat{\mathbf{s}}_{meas} = \mathbf{R}(\bar{\mathbf{q}}) \begin{bmatrix} \cos el \cos az \\ \cos el \sin az \\ \sin el \end{bmatrix}, \tag{8}$$

and the measured azimuth and elevation are taken in the ENU triad built on $\hat{\mathbf{u}}_{meas}$; the predicted angles of each particle follow from the ephemeris Sun position in that particle's own ENU frame, so that the innovations ν^i_{az}, ν^i_{el} depend on its absolute latitude and longitude.

Weighting and resampling. The active channels are combined into $D^i = \boldsymbol{\nu}^{i\top}\mathbf{R}^{-1}\boldsymbol{\nu}^i$, with $\mathbf{R} = \mathrm{diag}(\sigma^2_{bar}, \sigma^2_{alt}, \sigma^2_{az}, \sigma^2_{el})$, and weighted by a heavy-tailed, tempered Student's-t likelihood,

$$\log w^i_k = \log w^i_{k-1} - \beta\,\frac{\nu_t + m}{2}\log\left(1 + \frac{D^i}{\nu_t}\right), \tag{9}$$

with ν_t the degrees of freedom, β the tempering exponent and m the number of active channels. The heavy tail keeps an occasional anomalous match from killing a particle. The tempering $\beta < 1$ makes each update only a small correction of the weights, so that a particle is judged on the accumulated evidence of many successive epochs rather than on a single one: the weights then reflect how well the particle matches the terrain profile travelled over a long stretch of trajectory, which carries much more topographic information than a point-by-point comparison. Since the residuals are correlated over hundreds of seconds while the update runs at 10 Hz, an untempered likelihood would also treat correlated samples as independent and collapse the swarm within seconds. Degeneracy is measured by the effective sample size [33, 23],

$$N_{eff} = \frac{1}{\sum_{i=1}^{N}(w^i_k)^2}, \tag{10}$$

which equals N for uniform weights and 1 when a single particle carries them all. When $N_{eff} < 0.5N$ the swarm is regenerated by systematic resampling [34], the weights are reset to $1/N$, and each position is roughened [22] by a Gaussian displacement of 0.05° to counteract sample impoverishment. The estimate is a weighted mean over the particles within 25 km of the most probable one, while its covariance $\boldsymbol{\Sigma}_p$ is computed over the whole swarm.

Remark on the longitude problem and the role of the Sun sensor. With the barometer and the altimeter alone, the estimate can drift over flat terrain and become trapped in a local minimum of the topography (Fig. 7): four

runs out of ten diverge in this way, through a longitude lock that is never corrected. The Sun channels, which have no local minima and depend on absolute longitude, are the only measurement able to detect and correct such a divergence.

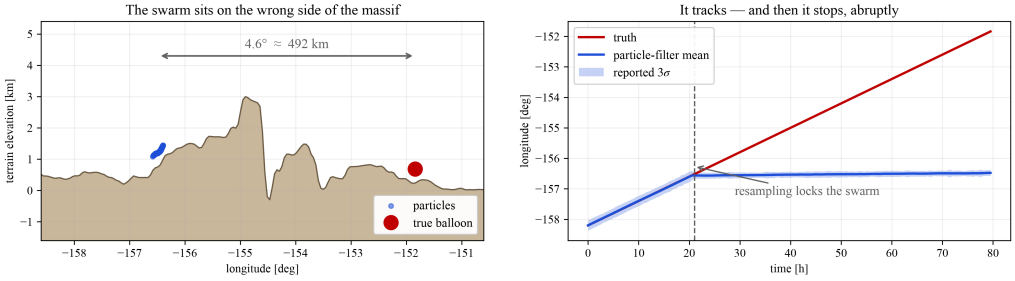


Figure 7: Terrain trapping with altimeter and barometer only: the estimate stays locked on one side of a massif (4 runs out of 10 diverge), motivating the Sun channel.

4.4. Uncertainty Exchange Between the Two Filters

In the cascade, point estimates and their uncertainties cross the interface (Table 4). The uncertainty of each transferred quantity is added to the receiving filter's measurement noise, recomputed at every epoch.

Table 4: Quantities exchanged between the two estimators at each epoch. Three of the five entries are uncertainties rather than values.

Quantity	Direction	Purpose in the receiving filter
$\bar{q}_k$	MEKF $\rightarrow$ PF	tilts the altimeter slant range; builds the measured horizon frame
$\bar{b}_{a,k}$	MEKF $\rightarrow$ PF	de-biases the specific force before the vertical is reconstructed
$\sigma_{\theta,k}$	MEKF $\rightarrow$ PF	inflates the Sun-channel noise
$\hat{p}_k$	PF $\rightarrow$ MEKF	builds the local-vertical and Sun references
$\sigma_{p,k}$	PF $\rightarrow$ MEKF	inflates the accelerometer and Sun measurement noise

Attitude uncertainty into the particle filter. For particle i, the attitude covariance $\mathbf{P}_{\theta\theta}$ is projected on its own local vertical $\hat{\mathbf{u}}_B^{(i)} = \mathbf{R}(\bar{\mathbf{q}})^\top \hat{\mathbf{u}}(\mathbf{p}^i)$ to give the yaw uncertainty, the rest of the trace being attributed to the two tilt axes together with the accelerometer noise. Since a tilt moves the elevation one-for-one and the azimuth by $1/\cos(el)$, while a yaw moves only the azimuth,

the Sun channels use

$$\sigma_{el} = \sqrt{\sigma_s^2 + v_{tilt}}, \qquad \sigma_{az} = \sqrt{\sigma_s^2 + \sigma_\psi^2 + \frac{v_{tilt}}{\cos^2 el}}, \tag{11}$$

$$\sigma_\psi^2 = \hat{\mathbf{u}}_B^{(i)\top} \mathbf{P}_{\theta\theta} \hat{\mathbf{u}}_B^{(i)}, \quad v_{tilt} = \frac{\operatorname{tr} \mathbf{P}_{\theta\theta} - \sigma_\psi^2}{2} + \frac{\sigma_w^2}{g_0^2}. \tag{12}$$

Position uncertainty into the MEKF. The swarm covariance is converted into the horizontal arc length $\sigma_p^2 = (r\sigma_\varphi)^2 + (r\cos\varphi\,\sigma_\lambda)^2$, with $r = R_V + h \approx 6107\,\mathrm{km}$. For the Sun sensor it produces a parallax $\sigma_{par} = (180/\pi)\,\sigma_p/d_\odot$ degrees, with $d_\odot \approx 1.08 \times 10^8\,\mathrm{km}$; for the accelerometer it tilts the local vertical by $\sigma_u = \sigma_p/(\sqrt{2}\,r)$ per horizontal axis. The two covariances become

$$\mathbf{R}_s = \operatorname{diag}\left(\sigma_s^2 + \frac{\sigma_{par}^2}{\cos^2 el},\ \sigma_s^2 + \sigma_{par}^2\right), \tag{13}$$

$$\mathbf{R}_a = \operatorname{diag}(\sigma_w^2 + \sigma_{k,i}^2) + (g_0\sigma_u)^2\left(\mathbf{I}_3 - \hat{\mathbf{g}}\hat{\mathbf{g}}^\top\right), \tag{14}$$

the tilt term being injected in rank-two form so that the radial channel, which carries the accelerometer-bias information, is untouched. The local vertical is $r/d_\odot \approx 1.8 \times 10^4$ times more sensitive to a position error than the Sun direction, and the position error is correlated in time: the MEKF therefore corrects with the Sun sensor every 10 s and with the accelerometer only every 1000 s.

5. Simulations and Results

5.1. Simulation Setup

After a reference run (Section 5.2), the estimator is therefore assessed through Monte Carlo campaigns, organised as a sensitivity analysis on the two main assumptions: the knowledge of the atmosphere (Section 5.3) and the accuracy of the Sun sensor (Section 5.4).

Each configuration is simulated over one hundred independent runs. Each run starts at the epoch of Table 2, with the balloon at 58 km of altitude and 0° of longitude, a latitude drawn uniformly between −20° and 20°, and its own atmospheric-perturbation and sensor-noise seeds (Fig. 8). Each run simulates 100 h of flight at $\Delta t = 0.1\,\mathrm{s}$, within which the far-side window lasts about 65 h, the Sun setting about 46 h after the link is lost. The estimator is the full architecture of Section 4, with $N = 1000$ particles and the cadences of Section 4. The initial attitude of the MEKF is computed with the TRIAD method [35] from the first Sun-sensor and accelerometer readings. The MEKF is started 2 h before the blackout, while the position

is still known, so that its initial transient has settled when the particle filter takes over; the particles are initialised about the last known state with $\sigma_\varphi = \sigma_\lambda = 0.1^\circ$, $\sigma_h = 50\,\mathrm{m}$ and $\sigma_v = 1\,\mathrm{m\,s^{-1}}$.

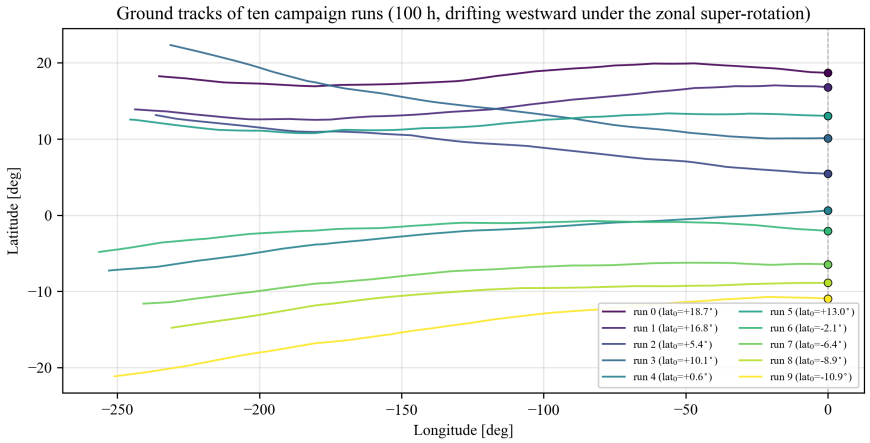


Figure 8: Ground tracks of 10 Monte Carlo runs, launched at longitude 0° with latitude uniform in $[-20^\circ, 20^\circ]$.

Every run is scored over the sunlit arc only, on three statistics:

- **Lost runs**, whose position error crosses a 500 km threshold before sunset and does not recover, counted out of one hundred;
- **RMSE**, the position root-mean-square error over the sunlit arc, averaged over the runs that are not lost;
- 3σ **coverage**, the fraction of sunlit epochs at which the true position error lies within the filter's own 3σ bound, averaged over the runs that are not lost.

5.2. Reference Run

Before the campaigns, the coupled estimator is tested on a single reference run. Figure 9 zooms into a ten-minute window late in the sunlit arc: roll and pitch oscillate by a few tenths of a degree as the balloon swings, and the MEKF follows that oscillation with the true value inside its declared 3σ band; yaw, driven by the much larger and slower rotation of the heading, is reproduced with no visible lag.

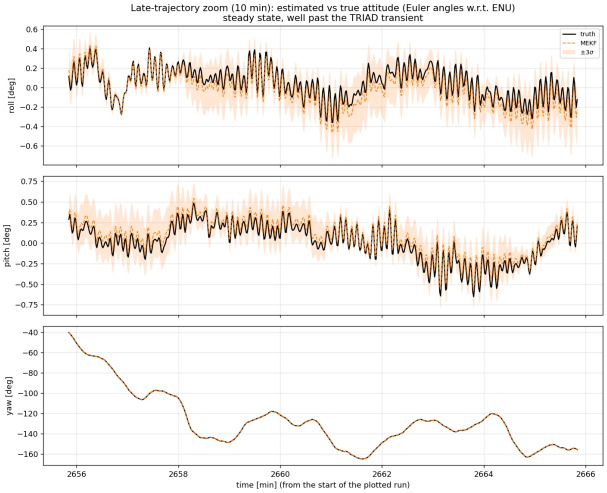


Figure 9: Single run: estimated vs. true roll, pitch and yaw over a 10 min window, with the MEKF $\pm 3\sigma$ band.

The particle-filter mean follows the true latitude and longitude closely, and the altitude estimate follows the oscillations of the balloon around its float level (Fig. 10). Figure 11 quantifies the resulting accuracy: a horizontal RMSE of 23.7 km, with a worst-case excursion of about 80 km, and an error that stays within the filter's own 3σ bound.

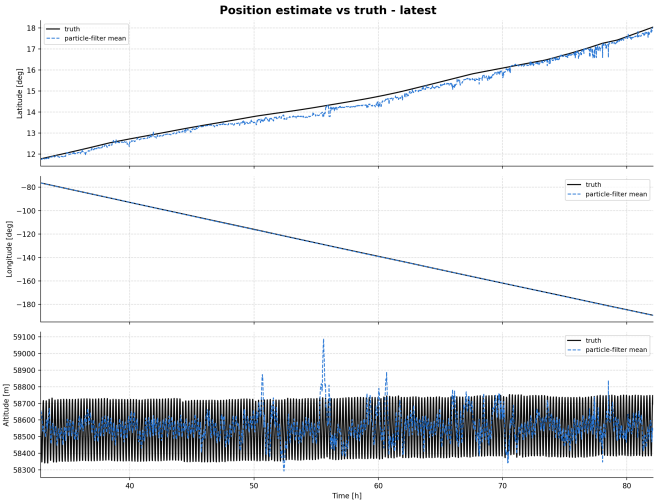


Figure 10: Single run: particle-filter position mean vs. truth (latitude, longitude, altitude).

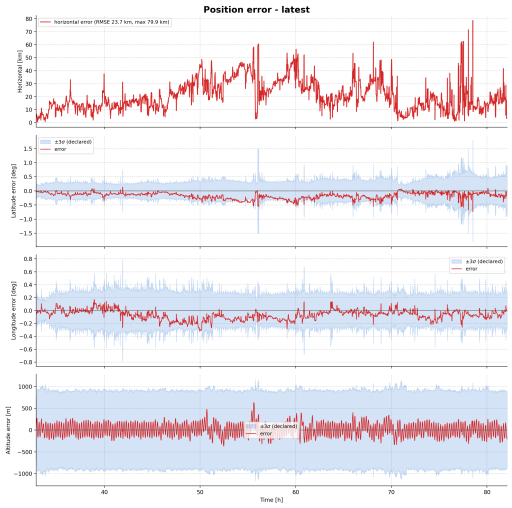


Figure 11: Single run: horizontal position error (RMSE 23.7 km, maximum about 80 km) and per-axis error against the declared $\pm 3\sigma$.

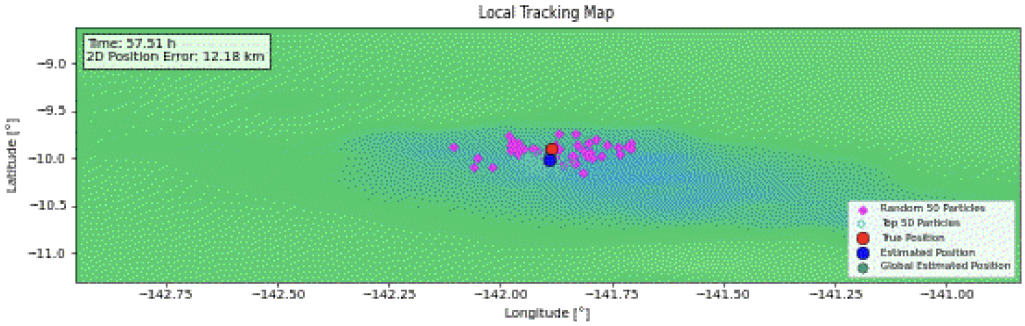


Figure 12: Particle swarm over the local topography at $t = 57.5$ h, with true and estimated positions.

5.3. Sensitivity to the Atmospheric Model

Every result of this work already includes a persistent modelling error, since the particles fly the mean fields of the Venus Climate Database while the truth flies its perturbed version. This campaign degrades the onboard atmosphere further, by reading the same database five Venus hours out of phase. Since the database is indexed by local solar time, this corrupts the filter's knowledge of the diurnal cycle of the atmosphere.

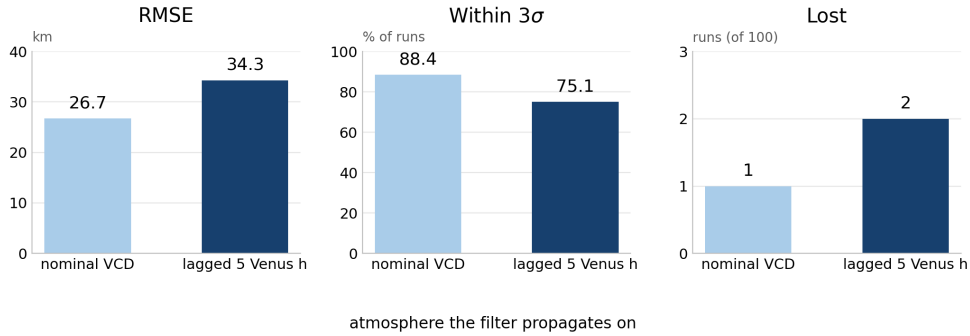


Figure 13: Atmosphere campaign (100 runs): reading the database five Venusian hours out of phase costs 7.6 km of RMSE (26.7 to 34.3 km).

Table 5: Sensitivity to the fidelity of the filter's onboard atmosphere. One hundred runs per row, sunlit arc only, identical seeds and initial latitudes.

Atmosphere the filter propagates on	RMSE [km]	3σ coverage [%]	Lost
Nominal database	26.7	88.4	1/100
Database lagged 5 Venus h	34.3	75.1	2/100

The estimator is sensitive to the knowledge of the atmosphere (Table 5, Fig. 13): the lagged database worsens the RMSE by 7.6 km and lowers the 3σ coverage by thirteen percentage points, since the particles are propagated on a wind that departs further from the real one. The RMSE nonetheless remains within the 50 km requirement. The number of lost runs, one or two out of one hundred, against four out of ten in the configuration without the Sun channels in the position update (Section 4), reflects the importance of the double use of the Sun sensor, in both the attitude and the position filter. A way to reduce the gap between the modelled and the real atmosphere is proposed in Section 6.

5.4. Sensitivity to the Sun Sensor Noise

The Sun sensor has not been designed in detail, and the baseline $\sigma_s = 1 \times 10^{-4\circ}$ may not be achievable through the cloud deck. This campaign repeats the protocol over three decades of σ_s with the nominal atmosphere model.

Table 6: Sensitivity to the Sun-sensor noise σ_s. One hundred runs per row, sunlit arc only, nominal atmosphere throughout.

σ_s	RMSE [km]	3σ coverage [%]	Lost
1×10^{-4} deg	26.7	88.4	1/100
1×10^{-2} deg	27.4	91.8	2/100
1×10^{-1} deg	25.7	95.3	1/100

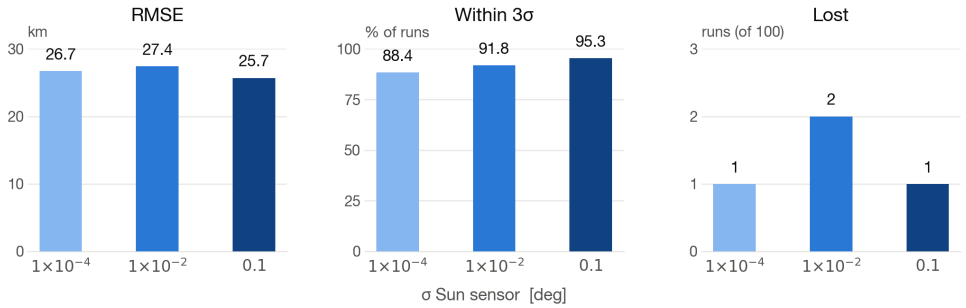


Figure 14: Sun-sensor campaign (100 runs per row): across three decades of noise the RMSE moves by only 1.7 km.

Across three decades of sensor noise the results do not change (Table 6, Fig. 14): the RMSE stays between 25.7 km and 27.4 km, a spread of 1.7 km that is not even monotonic in σ_s, and the number of lost runs stays at one or two out of a hundred. The estimate is always carried mainly by the altimeter and the barometer, which constrain the position at a much finer horizontal scale than the Sun direction: a difference of a fraction of a degree in Sun azimuth or elevation corresponds to hundreds of kilometres between two particles. The Sun channels are used only to detect and correct a divergence, when the estimate is locked in a local topographic minimum. For this reason the RMSE stays at the level set by the terrain-relative channels, of a few tens of kilometres, and does not deteriorate with a worse Sun sensor.

The 3σ coverage, instead, increases with σ_s. This is not a real improvement in the consistency of the filter: a larger sensor noise makes the measurement covariance $\mathbf{R}$ of the particle filter larger, so the weights discriminate less between particles and the swarm stays more dispersed, which inflates the covariance it reports.

6. Conclusions and Future Work

6.1. Main Findings

6.2. Limitations and Future Work

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Appendix A. Additional Figures

Appendix A.1. Background and Environment



Figure A.15: Venus seen by Mariner 10: a global, unbroken deck of sulfuric-acid clouds hides the surface from visible-wavelength instruments (NASA/JPL-Caltech).

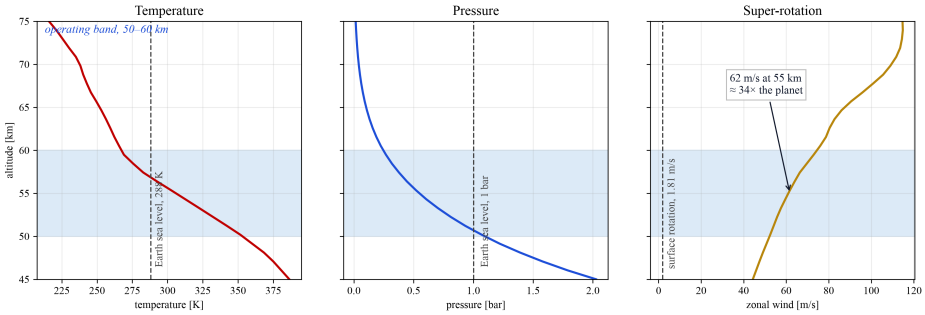


Figure A.16: Temperature, pressure and zonal wind between 45 and 75 km (Venus Climate Database). At 50–60 km conditions approach Earth's, with a zonal wind of about 62 m/s.

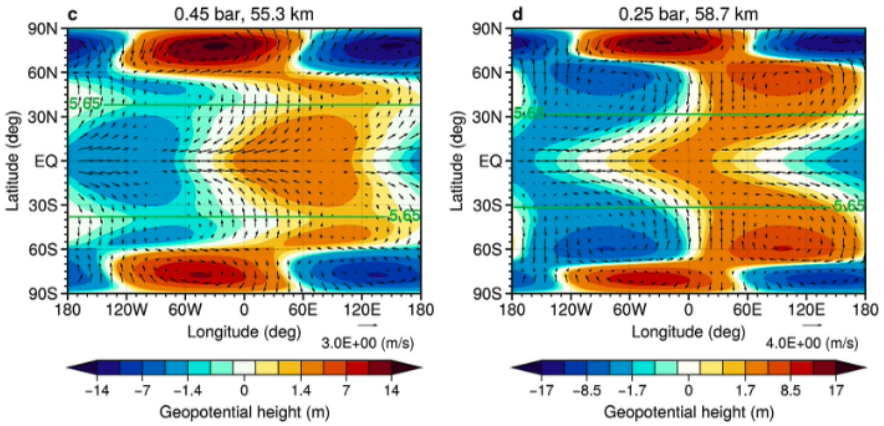


Figure A.17: Planetary-scale Rossby–Kelvin wave: wavenumber-one geopotential anomaly sweeping around the planet with a 5.65-day period (Lai et al.).

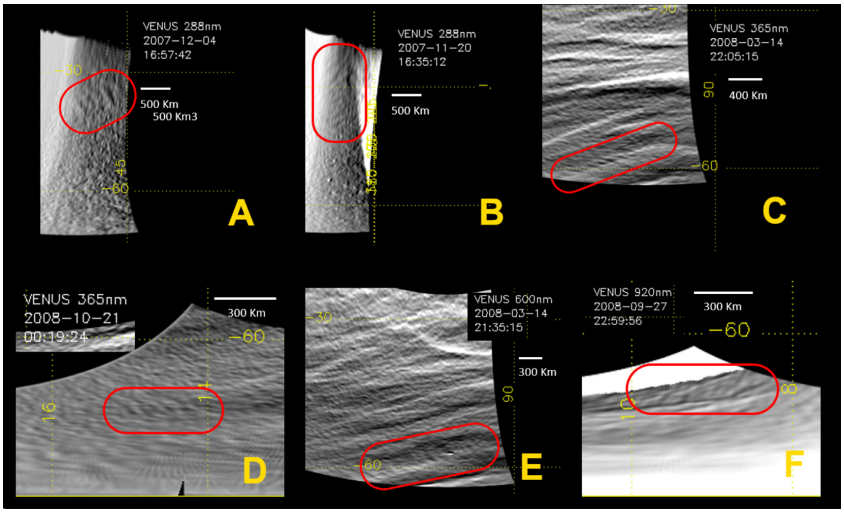


Figure A.18: Meso-scale gravity-wave packets in VIRTIS-M cloud imagery, with wavelengths of tens of km and not tied to any topography (Silva et al.).

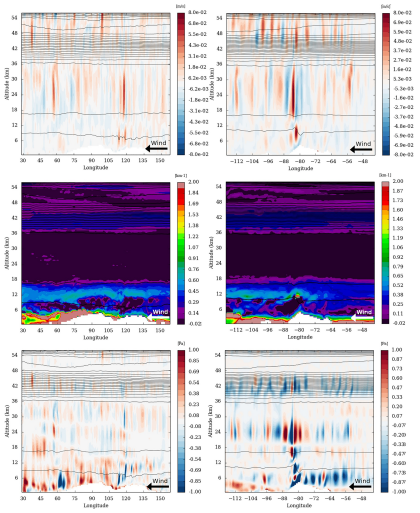


Figure A.19: Orographic waves: longitude–altitude cross-sections of vertical wind, Scorer parameter and pressure perturbation, anchored to the topography (Lefèvre et al.).

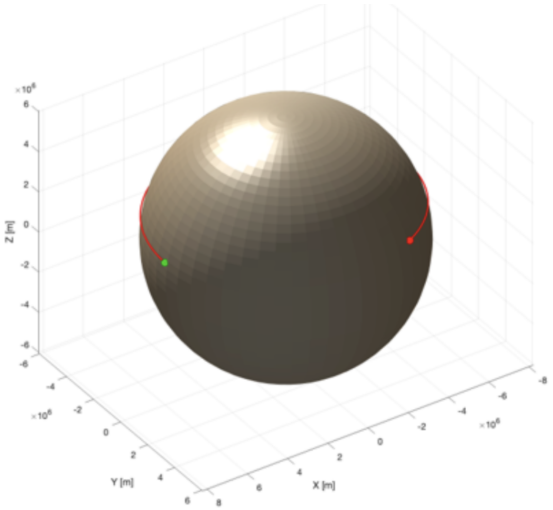


Figure A.20: Ground-truth trajectory over the 100 h reference run in VCVF coordinates (green: start, red: end).

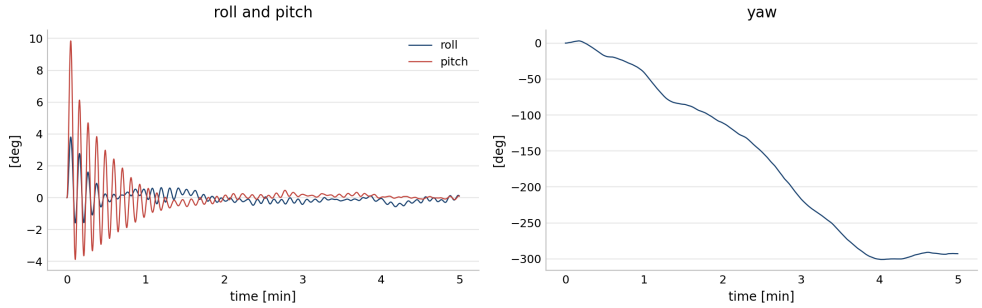


Figure A.21: Truth attitude over five minutes. Left: roll and pitch (damped pendulum, within about 1° of the vertical). Right: yaw, which has no restoring moment and drifts freely.

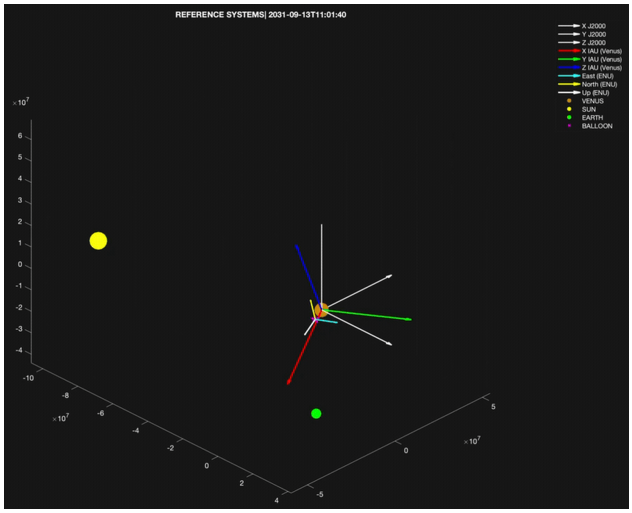


Figure A.22: Space environment at the start epoch from SPICE: positions of Sun, Earth and balloon relative to Venus, with J2000, IAU_VENUS and ENU axes.

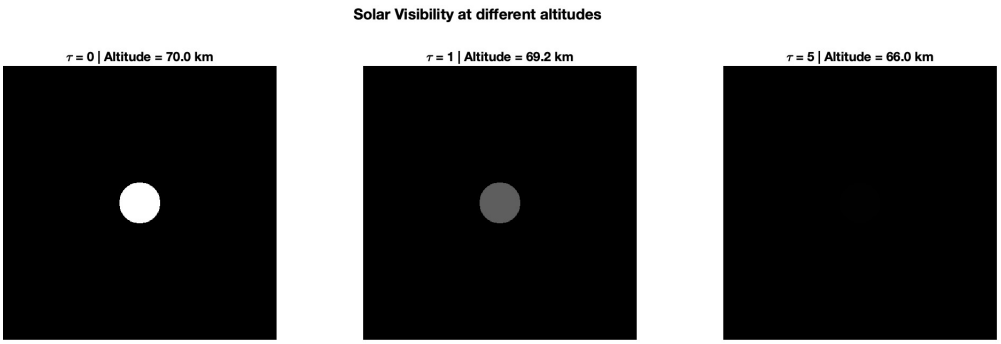


Figure A.23: Solar disk seen from below the cloud deck at 70, 69.2 and 66 km: the direct beam is invisible at 66 km, so an optical Sun sensor is not feasible.

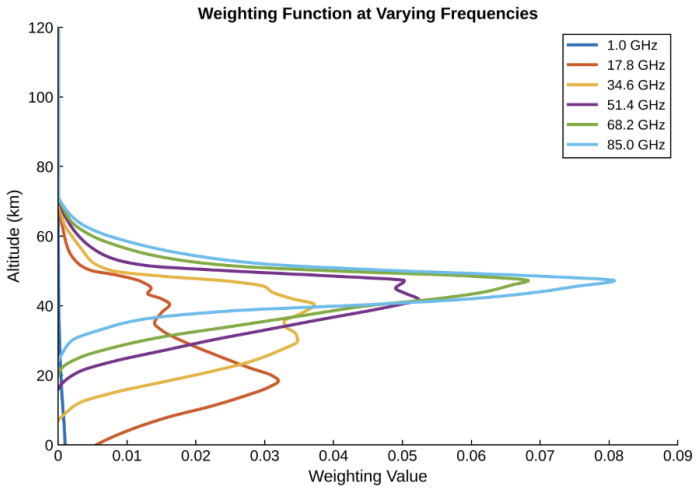


Figure A.24: Weighting functions of the Venus atmosphere between 1 and 85 GHz: below about 18 GHz the atmosphere above the balloon is transparent to the Sun, which motivates a microwave Sun sensor.

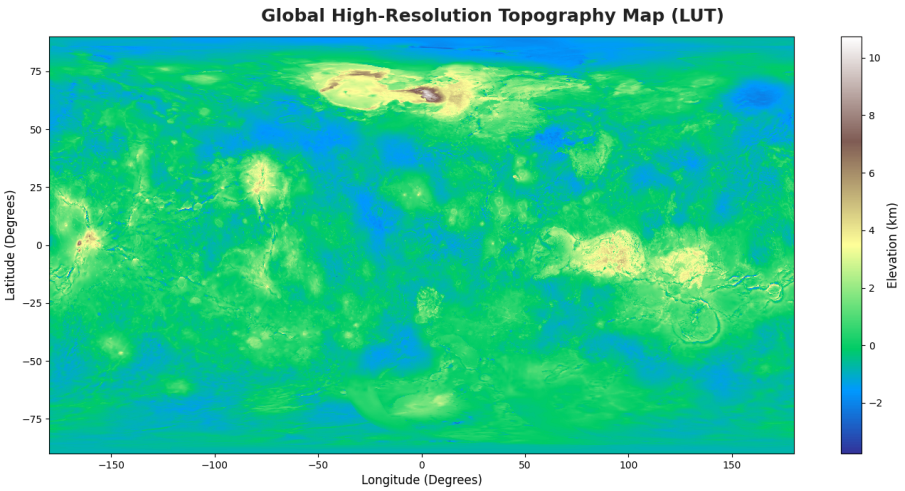


Figure A.25: Magellan global topography used as the altimeter look-up table. Aphrodite Terra and Ishtar Terra make the altimeter informative; the plains are where it is weakest.

Appendix A.3. Additional Reference-Run Results

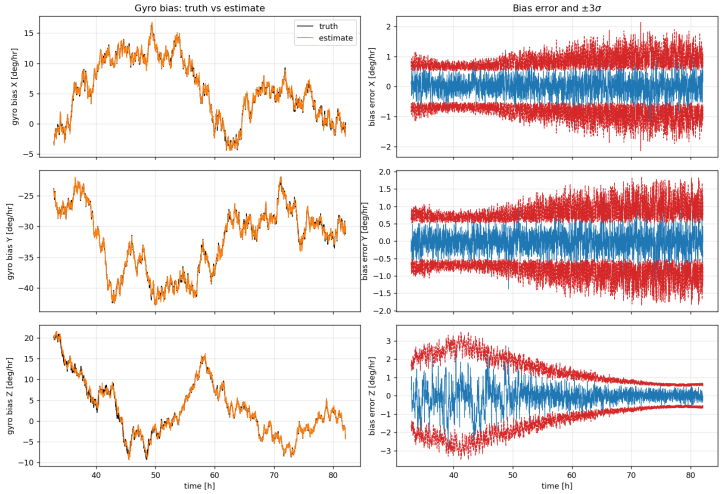


Figure A.26: Single run: gyroscope bias, truth vs. MEKF estimate (left) and error against the declared $\pm 3\sigma$ (right).

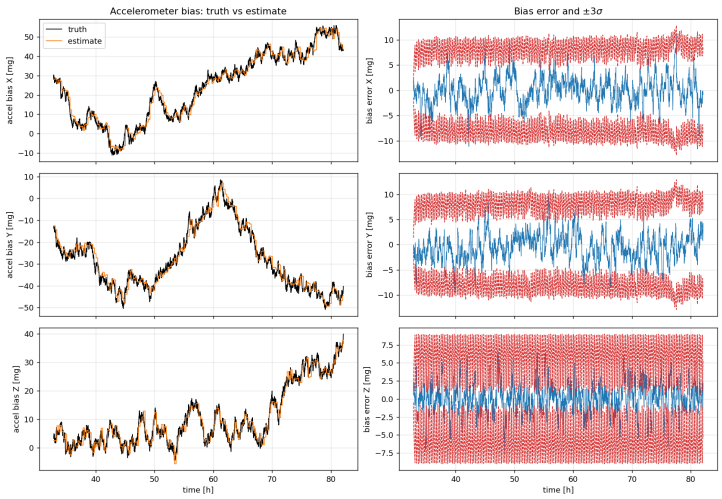


Figure A.27: Single run: accelerometer bias, truth vs. MEKF estimate (left) and error against the declared $\pm 3\sigma$ (right).

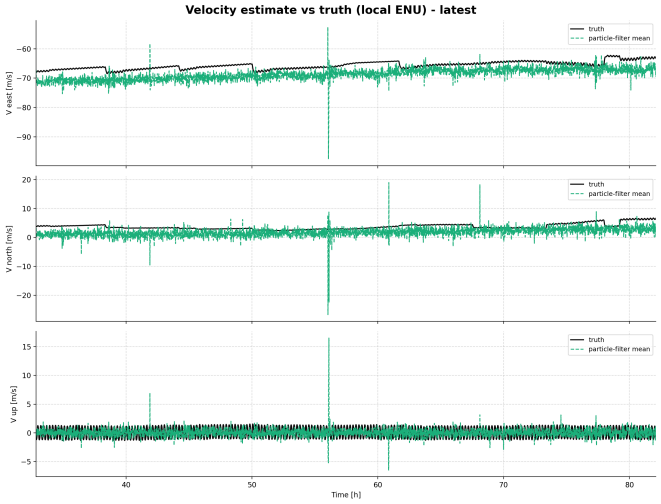


Figure A.28: Single run: particle-filter velocity mean vs. truth in the local ENU frame.

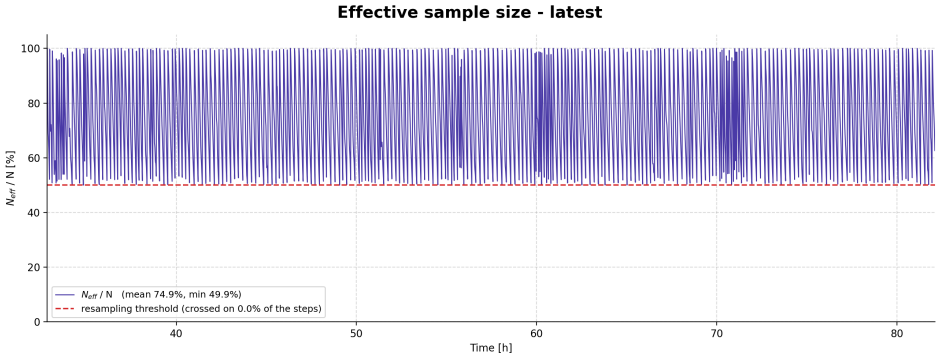


Figure A.29: Effective sample size N_{eff}/N over the run, with the resampling threshold.

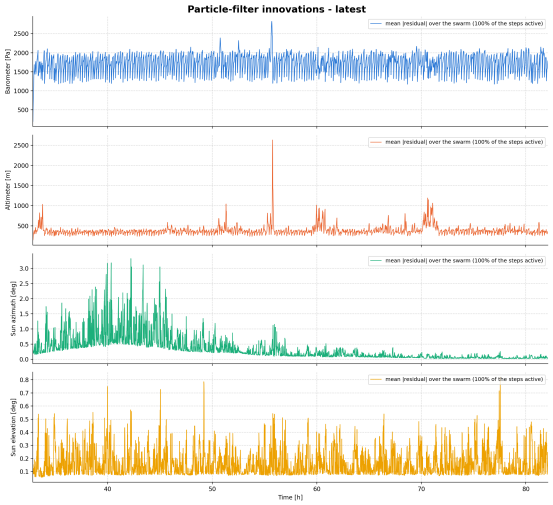


Figure A.30: Mean absolute residual over the swarm for the four particle-filter channels (barometer, altimeter, Sun azimuth, Sun elevation).



Figure A.31: Sensor usage over Sun-visible epochs: duty cycle, share of the particle filter's selection, and share over time.